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Level: Master's Programme of Studies

Network Security

Case 6

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## **Abstract**

Execution of a DNS Spoofing attack targeting the DNS server infrastructure implemented in the previous exercise. Using Ettercap on Kali Linux, a MITM position was established via ARP Poisoning. Subsequently, utilizing the dns\_spoof plugin, the victim's traffic was successfully redirected to a malicious webpage hosted on Apache server, bypassing the legitimate destination.

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## **Introduction**

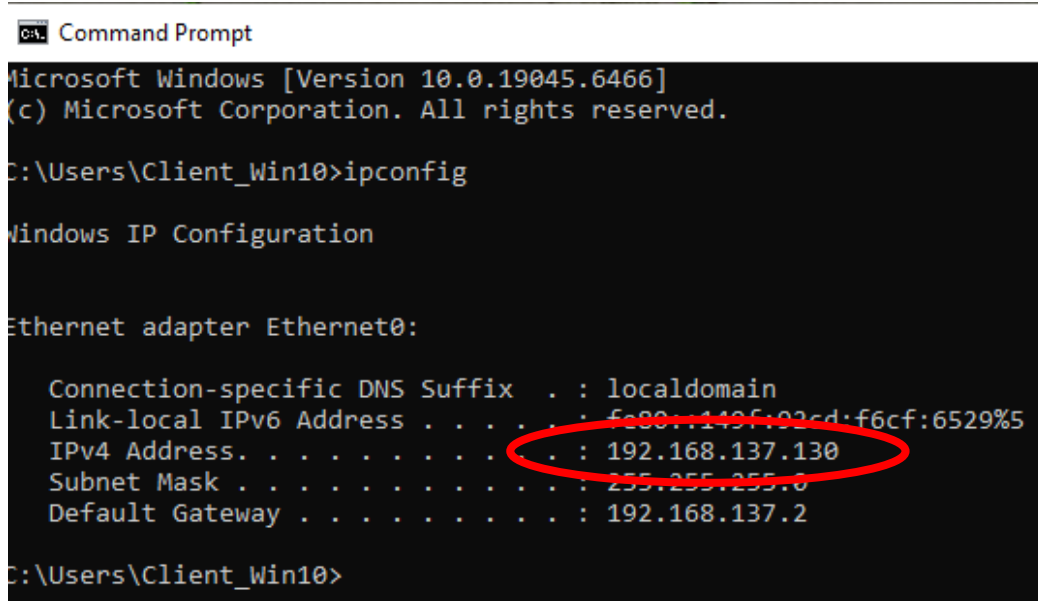
DNS (Domain Name System) is one of the cornerstones of the internet's operation, undertaking the mapping of website names (domain names) to IP addresses. Due to its critical importance, it is a frequent target of cyberattacks. One of the most widespread attack techniques is DNS Spoofing.

In this assignment we will implement such an attack, and as a basis we will use the infrastructure developed in the previous exercise (Case 5), where a legitimate DNS server (Ubuntu/BIND9) was installed and configured.

The purpose of the exercise is to understand the mechanisms of MITM attacks. We will use the Kali Linux operating system and the Ettercap tool to perform ARP Poisoning, interposing ourselves between the victim (Windows 10) and the DNS server. The goal is to lead the victim to a controlled page (kalaitzon.gr) hosted on the attacker's Apache Web Server.

## **Practical part of the assignment**

First, we begin by finding the IP addresses with the command `ip a` on Kali Linux and Ubuntu, and `ipconfig` for Windows 10. They are the same IPs from the previous exercise where we created the DNS Server.



```
CA: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Client_Win10>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : localdomain
    Link-local IPv6 Address . . . . . : fe80::149f:93cd:f6cf:6529%5
    IPv4 Address. . . . . : 192.168.137.130
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.137.2

C:\Users\Client_Win10>
```

**Attacker (Kali): 192.168.137.128**

**DNS Server (Ubuntu): 192.168.137.129**

**Victim (Windows 10): 192.168.137.130**

With the command `sudo systemctl start apache2` we turn the user (Kali Linux — the attacker) into a server so that it can host websites. We do this in order to 'serve' the page edited by us to the victim. The page will bear only the title 'YOU HAVE BEEN HACKED'. For this reason we use `apache2`. With the next command (`sudo nano /var/www/html/index.html`) we open the default root directory where `apache` looks for the website's files. We observe the following image when we open the file:

We will delete all the content and leave only the `<h1> YOU HAVE BEEN HACKED!`  
`</h1>`. It is not a necessary step, but through it we show a tangible result of this attack.

We save it by pressing `Ctrl+O` and `Enter` and exit the document by pressing `Ctrl+X`. To check that the page works, we will type into our browser the IP `127.0.0.1`. This address is the most famous IP and is essentially the localhost. To check whether the `apache` we

just set up on the attacker works correctly, we ran the webpage on the same host, without going out onto the network. We observe that the webpage is functional and it remains to proceed to the next part: Ettercap

With the command `sudo nano /etc/ettercap/etter.dns` we open the configuration file of `dns_spoof`, the Ettercap tool's plugin. In other words, `etter.dns` contains the domains we want to 'break' and send to our victim.

After opening the file, we will write the following, both mandatory and optional:

- **kalaitzon.gr:** This line catches the case where someone types the plain name, with nothing in front
- **\*.kalaitzon.gr:** Means 'anything' before `kalaitzon.gr`. E.g. if the user types `mail.kalaitzon.gr`, `admin.kalaitzon.gr` or anything else
- **www.kalaitzon.gr:** This is the classic case where we type `www`. Normally the previous case covers it, but for certainty we declare it separately **www.kalaitzon.gr**

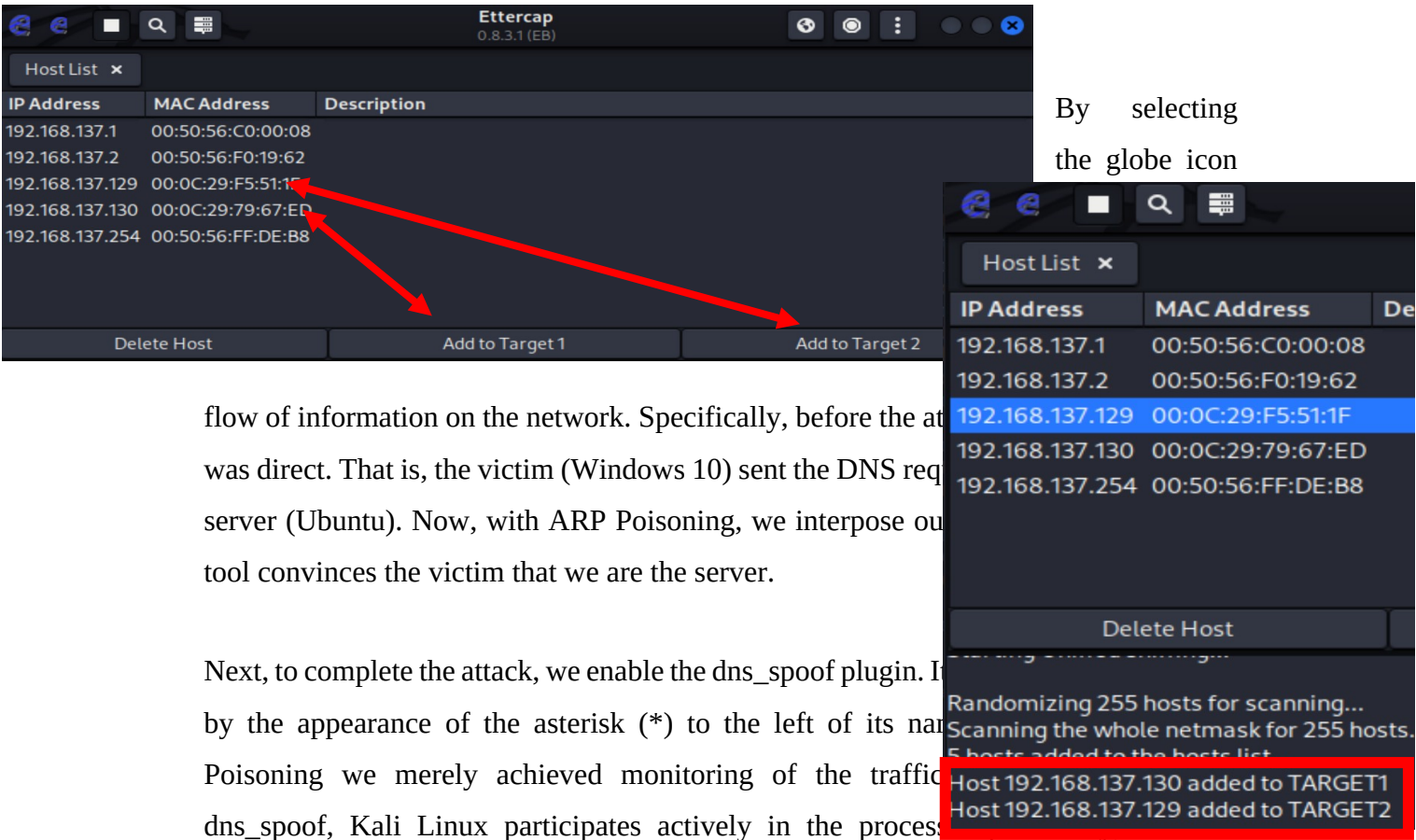
In this way we specify that if some other user on the network asks for the website `kalaitzon.gr`, for example, then instead of feeding it back to the Ubuntu IP as the DNS server, we feed it back to the attacker's IP (Kali Linux) without the victim knowing, until, of course, they see the message on their screen ('YOU HAVE BEEN HACKED!').

Finally, we give administrator (root) privileges in order to monitor the network as a MITM with the Ettercap tool. In addition, with `-G` we open a graphical interface to carry out the necessary actions this way, instead of typing them. In a separate window, then, we observe that the network interface is correctly set to the Kali Linux network card, and for this reason we proceed normally to the next step.

With the option indicated by the arrow, Ettercap instructed the network card (eth0) to enter Promiscuous Mode. That is, from reading only the packets destined for this particular host (Kali Linux), the network card now reads whatever moves. In other words, it monitors the network traffic.

By selecting this icon, we see that the scanning process was completed and it displays all the hosts that Ettercap detected on the network. We see the IP of both the victim and the DNS server.

We will select our targets, specifically Windows 10 as target1 and the Ubuntu DNS server as target2, as shown in the image below.



flow of information on the network. Specifically, before the attack was direct. That is, the victim (Windows 10) sent the DNS request to the DNS server (Ubuntu). Now, with ARP Poisoning, we interpose our tool convinces the victim that we are the server.

Next, to complete the attack, we enable the dns\_spoof plugin. It is indicated by the appearance of the asterisk (\*) to the left of its name. With ARP Poisoning we merely achieved monitoring of the traffic. With dns\_spoof, Kali Linux participates actively in the process. It no longer monitors the flow of packets and, as soon as it detects a DNS request concerning the



domain kalaitzon.gr, it intervenes immediately. Instead of allowing the response from the legitimate server, it sends the victim a spoofed reply with the attacker's IP (.128), thus redirecting the traffic.

Finally, to verify the attack, we go to the victim's computer (Windows 10) and open the browser. Typing the address `www.kalaitzon.gr`, we observe that the page with the message 'YOU HAVE BEEN HACKED!' appears. This fact confirms that the DNS Spoofing attack worked. Specifically, when the victim searched for the domain, the attacker (Kali Linux) intervened in the process. Through Ettercap, it sent a spoofed response before the legitimate DNS server could respond. As a result, the victim's browser was incorrectly routed to the attacker's IP (.128), loading the malicious webpage hosted on Kali's Apache Web Server, instead of the real content.

Additionally, to confirm the attack at the network level, the command `ping kalaitzon.gr` was executed in the victim's command line (CMD). As shown in the image, the system resolves the domain name to the IP address 192.168.137.128, which corresponds to the attacker's machine (Kali Linux). If the attack had failed, the command would have returned the IP 192.168.137.129 of the legitimate DNS Server. This result proves that the DNS Spoofing process was successful, as the victim is incorrectly routed to the malicious node.

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